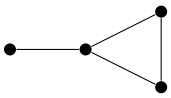
[GROUP NAME] **October 15, 2026**

Clarifier: Henry
Manager: Sharan
Reporter: Benjamin
Skeptic: Changhwan

Report: Today our tasks went well. We weren't able to complete 5 and 6, which was very unfortunate, so we better use our time wisely to compute more problems in the future. Personally question 9 had my interest for this session, ans it had shown a pseudo upper bound for a seemingly unexplainable reason. I would like to dive into this problem a bit more in the future. Nicely done report. If we are going to make this template available for future students, I'd need a short "how to use this template" guide I could give them. This is totally optional on your part.

SOLUTIONS & PROOFS:
Without Loss of Generality is abbreviated as WLoG within this document.

1. Find the chromatic polynomial of "the paw".



[Graph 1]

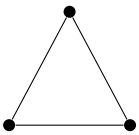
Solution. $p_k(\text{the paw}) = k(k-1)^2(k-2)$ Yes, but label the vertices here with appropriate values to show why!

2. Prove that if T_n is a tree on n vertices, $p_k(T_n) = k(k-1)^{n-1}$.

Proof. Intuitively, the root can be colored in k ways, and each of its child nodes can be colored in $k-1$ ways (excluding the root's color); subsequent child nodes can also be colored in $k-1$ ways (excluding their parent's color). Since a cycle would form if any child node were connected to a node other than its parent or its own children, the formula $p_k(T_n) = k(k-1)^{n-1}$ holds when the graph has a tree structure.Good! But what's a root? Try to define terms we haven't covered in class. □

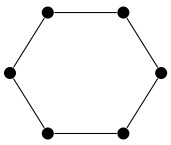
3. Prove by induction : $P_k(C_n) = (k-1)^n + (k-1)(-1)^n$.

Proof. Let n=3 be the base case. For n=3 $P_k(C_n) = k(k-1)(k-2)$ which goes well

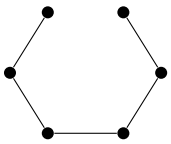


[Graph 2]

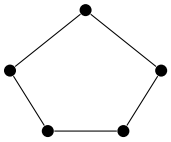
Assume it's true for n=x. For n=x+1 we can use the method we did during our class. The graphs below is an example for n=6. If we remove one edge we'll get the graph 4 or P_{x+1} . If we merge 2 vertices we can get the graph 5 or C_x. Since $P_k(P_{x+1}) = k(k-1)^x$, $P_k(C_x) = (k-1)^x + (k-1)(-1)^x$, $P_k(P_{x+1}) = P_k(P_{x+1}) - P_k(C_x) = (k-1)^{x+1} + (k-1)(-1)^{x+1}$You have a typo of P_{x+1} instead of C_{x+1}, and you may want to point out that the identity you're writing follows by factoring out a $(k-1)^n$. Also consider an align* environment for chains of equalities.



[Graph 3]



[Graph 4]



[Graph 5]

□

4. Let's think about cutthorat for a bit. We know that $G + G = P$.

   (a) Prove $P + P = P$

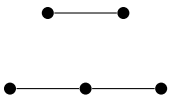
   Proof. Let the graphs be denoted by P_1 and P_2. WLoG, if player N plays on P_1, player P will also play on P_1 according to their strategy. Furthermore, if this process continues until all removable vertices in P_1 are eliminated, the players must then play on P_2; since P_2 is also of type P, player P will ultimately win. Therefore, $P + P = P$ holds. Good! □

   (b) Prove $P + N = N$

   Proof. If player N makes a move on an N-graph according to their winning strategy on their first turn, the graph transforms into a P-graph. Consequently, the roles of N and P are swapped, resulting in a situation equivalent to (a); thus, the relation $P + N = N$ holds.Good! □

   (c) Find an N graph and an N graph whose sum is a N graph.

   Solution. .



[Graph 6]

Good!

(d) Find an N graph and an N graph whose sum is a P graph.

Solution. .



[Graph 7]

Good!

5. Let G be a graph such that, for all vertices a and b, $\chi(G - \{a, b\}) = \chi(G) - 2$. Prove that G is a complete graph.

Solution. Our team has not been able to solve this question within time.

6. A graph G is called *color critical* if, for every vertex v, $\chi(G - \nu) < \chi(G)$. In other words, when you remove ν, the chromatic number goes down, and this is true for every vertex. Find a graph G such that G is not complete, $\chi(G) = 4$, and G is color critical.

Solution. Our team has not been able to solve this question within time. We don't expect every group to solve every problems!

7. Let us play COUPLES ARE FOREVER!

(a) Is COUPLES ARE FOREVER! starting with a pile of 9 coins N or P?

Solution. N.

Proof. Write N for the first player and P for the second.
The $T - T$(tweedledee tweedledum) theorem : All games of starting shape $2k$-path with $k > 1$ are N games.

Proof. Split the $2k$ path into k and k. Thereafter N mirrors every move of P onto the opposite half, so N always has a reply and moves last. □

Let the first player cut between the 6th and 7th positions.

$$[1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad 9]$$

$$[1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6] \; [7 \quad 8 \quad 9]$$

Since a live 3-pile admits exactly one move (into 2/1) and is dead afterwards, a position whose only live piles are k threes has exactly k forced moves remaining, so the player to move wins if and only if k is odd.
The second player must cut either the 6-pile or the 3-pile, giving four cases.

i. The second player divides the 6-path into 1/5.
N answers by dividing the 5-path into 2/3, leaving two live 3-piles. Exactly two forced moves remain, so N moves last and N wins.

ii. The second player divides the 6-path into 2/4.
N answers by dividing the 4-path into 1/3, leaving two live 3-piles. Exactly two forced moves remain, so N moves last and N wins.

iii. The second player divides the 6-path into 3/3.
Every live pile is now a 3, and there are three of them. Exactly three forced moves remain, so N moves last and N wins.

iv. The second player divides the 3-path into 2/1.
The only live pile is a 6-path, so by the $T - T$ theorem N divides it into 3/3, leaving two live 3-piles. Exactly two forced moves remain, so N moves last and N wins.

Therefore N wins in every branch of the game tree, and this is an N game. Perfect! □

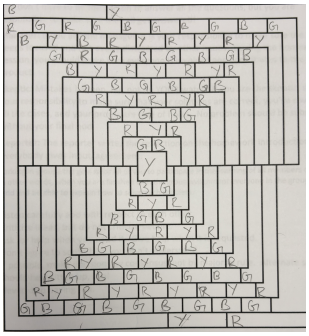
(b) Is COUPLES ARE FOREVER! starting with a pile of 100 coins N or P?

Solution. N.

Proof. Since $100 = 2(50)$ and $50 > 1$, the $T - T$ theorem applies directly and this is an N game. Good. □

8. Attempt to color the map on the cover of your folder in four colors.

Solution. .



[Graph 8]

Look at the tall regions; you have a ton of the same color touching each other!

9. Back to cutthroat, classify the path graphs up to p_{14}, or further.

Solution. We have confirmed that the known possible P-game values possible for p_n is $n = 1, 4, 12, 20, 30, 46, 72, 98, 124, 150, 176, 314, 408$. We have run a computer simulation with the given system. A naive solution stalls at $n = 55$, while the grundy solution continues without fault and is only limited by computational availability. Our simulations have proven that 408 is the largest n until 1M, and Caines et al. [1999](chain referenced from Guy and Nowakowski, "Unsolved Problems," 465.) has verified this value up to 50M. The existence of the 14th p-number is an unsolved problem. Nice work!

10. Prove that every non-empty graph's chromatic polynomial has at least one coefficient that is negative.

Proof. We prove this using two inductive steps. First, we induct on the order n, and second we induct on the size such that for all graphs of smaller order and size $n-1$ there exists a nonzero coefficient of k^{n-1}. You want the leading coefficient to be exactly one, not just nonzero, for this proof!The base case is the empty graph. Now we note that by the equation we found earlier, stating:

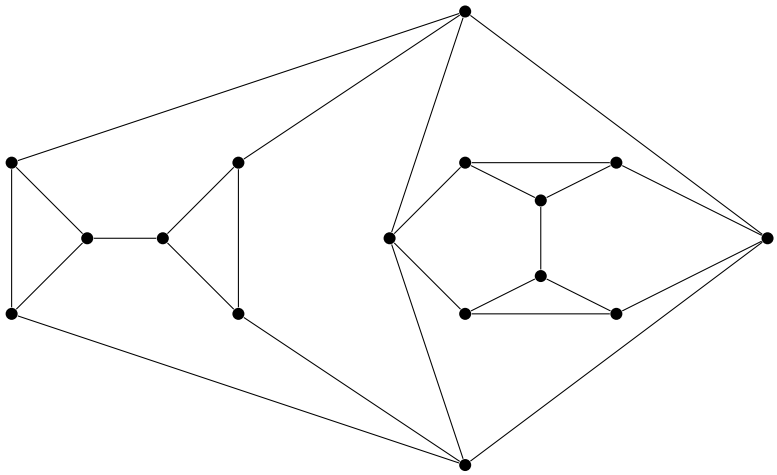
$$p_k(G) = p_k(G - e) - p_k(G * e),$$

We have $p_k(G * e)$ containing a k^{n-1} as it has one lower node and size satisfying our inductive assumption. Now we note that $p_k(G - e)$ maintains the same amount of nodes but one smaller size, but the extra node adds another factor of k to contain a k^n factor, which is unaffected by $p_k(G * e)$. Thus, $p_k(G)$ contains a k^n term. As k^n is the maximum value for $p_k(G)$ (the p_k value for an empty graph with n nodes), we must have some term that is subtracted to obtain a smaller value.This proof mostly works! You should mention why this maximum is only attained for the empty graph, but it's reasonably clear.

□

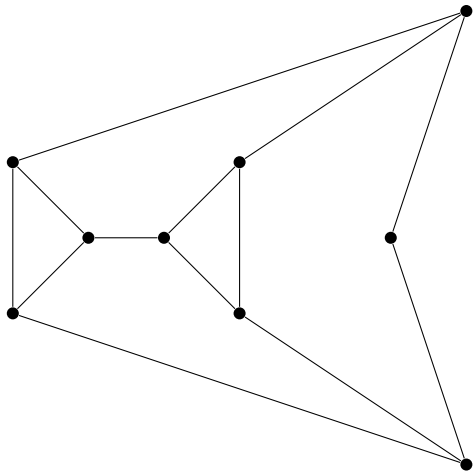
11. Prove that if a graph is color-critical with chromatic number k, the minimum degree of the graph has to be equal to $k-1$.

12. Prove that the graph below is color critical.



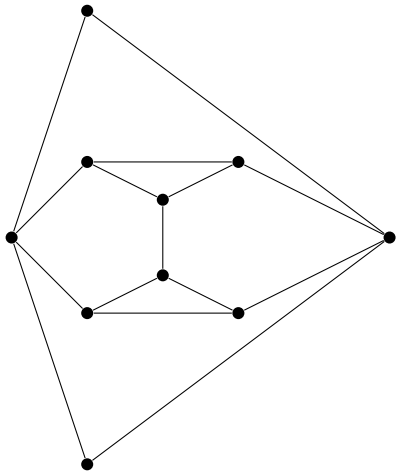
[Graph 9]

Proof. We aim to color using only three colors. Assuming the color of the middle point is 1, the topmost and bottommost points cannot be 1.



[Graph 10]

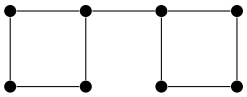
Case 1(Graph 10): If the topmost and bottommost points share the same color (e.g., 2), the top-left point becomes 1 and the bottom-left point becomes 3. Consequently, the middle-left point would have to be 2, making it impossible to fill all three points on the left side (top, middle, and bottom).



[Graph 11]

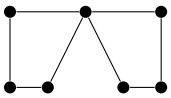
Case 2(Graph 11): If the topmost and bottommost points have different colors (top: 2, bottom: 3), the rightmost point becomes 1. Then, the upper-middle-right point must be 2, and the top-right point must be 3. Thus, the upper point among the middle-right positions becomes 1; following the same logic, the lower point among the middle-right positions must also be 1. This leads to a contradiction. Therefore, it is impossible to color using only three colors. Here you want to show this is 3-colorable after removing any vertex, not just that the original graph has chromatic number 4! □

13. Find the chromatic polynomial of the graph below.

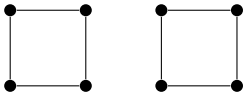


[Graph 12]

Solution. If we remove one edge from graph 9, it will look like graph 11. Additionally if we merge 2 vertices together it will look like graph 10. Since $p_k(graph10) = k(k-1)^2(k^2-3k+3)^2$, $p_k(graph11) = k^2(k-1)^2(k^2-3k+3)^2$, $p_k(graph9) = p_k(graph11)p_k(graph10) = k(k-1)^3(k^2-3K+3)^2$This is right, but you should briefly explain these chromatic polynomials. Also consider naming the graphs, maybe something like G, G_*, G_-. And inline math mode may not be best here.



[Graph 13]



[Graph 14]

14. *UNSOLVED* Is it true that, if, for all *adjacent* vertices a and b, $\chi(G - \{a, b\}) = \chi(G) - 2$ has to be complete?

15. *UNSOLVED* *Erdos : The Dense Triangle Free Conjecture.* If G is a triangle-free graph whose minimum degree exceeds $1/3$ of the number of vertices, the G can be colored in four colors. Can you prove it or find a counterexample?

16. *UNSOLVED* Are there more than 7 unipancyclic graphs?

Nicely done overall. Take Z's comments seriously, and be proud that you did a great assignment.